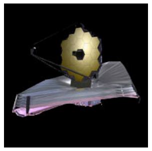

8369 - Making JWST as good as it should be: Bridging the NIRISS/AMI performance gap with model-based post processing

Cycle: 4, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Over the past two decades, direct imaging has emerged as a vital technique for studying exoplanets, complementing traditional radial velocity and transit methods. The James Webb Space Telescope (JWST) offers unprecedented sensitivity and Point-Spread-Function stability, enabling high-contrast imaging from 0.6 to 28 µm to detect lower-mass planets at wide separations. The Near Infrared Imager and Slitless Spectrograph with its Aperture Masking Interferometry (AMI) mode provides a unique opportunity to investigate gas giants at orbital separations around 3 AU—an underexplored region near the iceline where planet production is expected to peak. However, early AMI observations revealed degraded performance compared to predicted contrast due to systematic errors, necessitating improved calibration techniques. We propose to demonstrate a model-based, forward modeling approach that leverages wavefront sensing data on NIRISS archival data from commissioning, Early Release Science, and General Observer programs. By removing the need for reference star calibration, this image-plane approach is less sensitive to detector effects than techniques such as closure phase, addressing the dominant factor in AMI contrast performance while making observations significantly more time-efficient. The implications of this work extend beyond AMI observations, as the developed modeling techniques will enhance the accuracy of future

JWST programs and inform the design of next-generation space telescopes. By addressing the challenges of model-based techniques now, we will be in a much better position to inform upcoming missions, including the Roman Coronagraph and the Habitable Worlds Observatory.

OBSERVING DESCRIPTION